

Topographical Effects of Materials on Aerodynamics

The Aerodynamic Effect Created by Topography from GT3 Composite Materials

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Abstract:

This paper investigates how surface topography influences the aerodynamic performance of materials used in GT3 class racecars. Seven surface finishes representative of GT3 bodywork construction were examined: gloss carbon fiber, matte carbon fiber, gloss clear coat, matte clear coat, gloss vinyl wrap, matte vinyl wrap, and bare aluminum. Average surface roughness values (R_a) for each material were sourced from peer-reviewed literature and converted to equivalent sand-grain roughness (k_s) using the Nikuradse relation for use as wall roughness inputs in computational fluid dynamics simulations. Seven steady-state incompressible CFD simulations were conducted in SimScale using the $k-\omega$ SST turbulence model, with identical geometry, boundary conditions, and mesh resolution across all runs; only the wall roughness parameter was altered between simulations. The drag coefficient (C_d) and skin friction coefficient (C_f) were recorded for each material, along with the location of boundary layer transition. Results showed a strong positive relationship between surface roughness and drag, with C_d increasing by 124% from gloss carbon fiber ($C_d = 0.021$) to matte vinyl wrap ($C_d = 0.047$), and a clear performance threshold separating gloss finishes from matte finishes. These findings confirm that surface topography is a meaningful aerodynamic design variable for GT3 vehicles and show that smooth-finished carbon fiber composites offer the lowest drag among materials tested.

Introduction & Literature Review:

The aerodynamic design and performance of racecars in GT3 racing is based on efficiency and optimization since lost speed due to increased drag or imbalance can also lose racing time. This is why the material surface quality of different surfaces is crucial for aerodynamic engineers. The topography of manufactured surfaces have an effect on airflows that change based on the surface characteristics, the use of carbon fiber is the most common material choice for GT3 racing class where the finished products have a very smooth surface finish when properly manufactured and coated. This paper will present the impact of surface topography on aerodynamic performance of racecars and the relationship between surface quality and skin friction drag, and also show how surface roughness correlates with boundary layer formation, while also explaining why carbon fiber composites provide strong structural and aerodynamic advantages for designing a racecar.

The amount that air is displaced and how it behaves on the surface of a car is based on the boundary layer conditions of the vehicle (NASA, n.d.). When the surface of the vehicle is very smooth, the boundary layer will stay laminar for a longer distance, creating more controlled flow (NASA, n.d.). Once a surface becomes rough due to issues in texture, the boundary layer will change to turbulent flow. In turbulent flow over a rough surface the flow of the fluid loses more energy when it transfers momentum to the solid surface, causing an increase in drag forces that the racecar experiences. When a racecar's body has more texture or a matte finish, the texture increases near wall turbulence and can cause separation behavior depending on the pressure conditions that it experiences, needing more power from the engine in order to experience the same acceleration as a smoother finish. This is supported by work on a dimpled sphere, where the dimpled surface on a golf ball shows how controlled surface roughness can

change boundary layer behavior and affect drag characteristics (Benson, n.d.). GT3 vehicles are capable of operating under very high Reynolds numbers since their characteristics already exhibit large turbulent flow (De Groote, 2009). Due to this, introducing additional roughness to the vehicle will most likely increase overall drag forces and negatively affect performance.

The drag coefficient, also referred to as C_d , is a numerical value that accounts for all parts of drag that is made by the vehicle, including its shape, flow conditions, and surface. The drag equation, $D = \frac{1}{2}\rho V^2 A C_d$, changes C_d into the total drag for a specific shape at a specific speed. A change in the surface texture of a vehicle can impact the drag coefficient depending on what the surface roughness and finish characteristics are for the material, even if the shape of the shell stays the same, a very glossy smooth surface will have less total skin friction and, therefore, a lower C_d than a surface with coarse or pitted textures that create a greater skin friction. A smooth and waxed surface will have lower drag than a rough surface and this relationship is shown by the measured drag coefficients for various object shapes (NASA, n.d.). After research, aerodynamic engineers came to the conclusion that polishing and coating a panel of a racecar can provide a more efficient body and have less drag at higher speeds (Hatton, 2019). Other research shows that tests completed in a wind tunnel show that a shell with a consistent shape will have more drag forces if the surface is rough compared to smooth (Öztürk et al., 2020). Surface roughness influences drag due to differences in boundary-layer flow that develop as surfaces interact with the airflow created by a moving vehicle (NASA, n.d.).

One of the components of total aerodynamic drag is skin friction because of how the viscous shear of the air causes aerodynamic drag to act on the sides and top of the car's shell. Drag is defined as the aerodynamic force acting on the surface of an object due to the effects of pressure and viscous shear acting upon each point on the object's surface (Benson, n.d.). The

boundary layer is what is on the surface of the object when the moving air has interactions with the car's shell and the air molecules near the surface (NASA, n.d.). The smoother a finish, the less skin friction is generated. The opposite also applies and this should be taken into account when developing vehicles operating in the atmosphere (NASA, n.d.). Drag is affected heavily by skin friction when the vehicle is at high speeds and Reynolds numbers that are similar to that of racecars. Lowering the skin friction component of drag is possible when reducing surface roughness. This can happen by polishing the vehicle's panels and adding seals between the vehicle's seams. Keeping a smooth surface by using wax or laminar coatings can reduce the viscous interaction of the vehicle's surface with the atmosphere around it, which means it reduces skin friction drag (NASA, n.d.). Since racing teams are aware of this, they allocate resources to improve the finishes on the panels of the vehicles, because small changes in the topography can have measurable effects on speed and fuel efficiency.

Because of the smoothness of the surface, the boundary layer has a greater area where it can maintain a laminar flow, thus limiting energy loss. Surface roughness will cause disturbances in the air flow, causing an earlier change of the boundary layer from a laminar to a turbulent flow condition. Research by Öztürk et al. has shown that increased surface roughness constantly promotes a quicker change of the boundary layer to turbulent flow from laminar, resulting in an increase in the drag coefficient of the vehicle and reducing the aerodynamic efficiency (Öztürk et al., 2020). This is particularly significant in GT3 racing since vehicles typically operate in predominantly turbulent flow conditions due to the speed and shape of the vehicle. The introduction of additional surface roughness in this situation does not create any positive flow-tripping effect; it merely exacerbates the existing drag on the vehicle.

In addition to frictional losses, surface roughness affects the location of flow separation on the vehicle's exterior. Zdravkovich's work on rough-surface drag shows that roughness patterns can alter the location at which the airflow separates from the surface due to swift or inconsistent separation leading to increased size of the low-pressure wake behind the vehicle and consequently the pressure drag imposed by the vehicle (Zdravkovich, n.d.). In racing scenarios, poor and/or unavailable flow separation results in not only increases in drag but also decreases in the predictability of aerodynamic downforce; therefore, poor flow separation leads to both an unreliable vehicle balance and consequently reduces driver control when travelling at high speeds. Therefore, managing boundary layer characteristics through proper surface quality is not only a drag consideration but is fundamental to the overall stability of the vehicle.

Features that are not visible to the human eye also contribute to the overall aerodynamic drag of a vehicle. The peaks and valleys of a material's surface create small pockets of disturbance within the boundary layer, generating tiny frictional forces that accumulate across the entire surface of the vehicle. Jiménez et al. show that distributed roughness elements significantly alter transitional flow behavior and drag characteristics, confirming that surface texture is an important variable that must be carefully controlled in aerodynamic optimization (Jiménez et al., 2023). Further, research published in MDPI's Applied Sciences has confirmed that rough textures produce higher surface friction and consequently higher total aerodynamic loads than smooth surfaces, meaning that there are direct, quantifiable performance consequences resulting from material finishing choices ("Effect of Surface Roughness on Aerodynamic Loads of Bluff Body," 2024).

In motorsport, applying treatments and sealing composite material surfaces is normal as these techniques reduce the impacts of invisible or unwanted surface area irregularities or

imperfections (Hatton, 2019). Composite panels are usually made with resin systems and/or coatings that are made to fill microscopic gaps and to create a more uniform surface area that is also more aerodynamically advantageous compared to the raw materials that were used before. Race teams can manufacture composite panels that have surface finishes that could never have been achieved with traditional materials, due to the development of composite fabrication processes, composite panel surface textures contain fewer irregularities than traditional materials and because of that, they create less skin friction drag (Airport Metals, 2020). Composite panel surface sealing treatments also add to keeping the structural integrity of the composite material that the surface is made of. Without proper treatments or processes, the composite panels can start to degrade during the intense race and start to create new surface defects. Due to this, the type of material used to manufacture the composite panel will be a major structural and aerodynamic design consideration for the race teams. Adding to this, using materials that naturally have characteristics that support smooth finishes, either due to their manufacturing process, grain structure, or ability to accept sealing or protective compounds, will give engineers the ability to better adjust their aerodynamic performance. Materials that are difficult to finish smoothly or ones that cannot stay together due to the thermal or mechanical stresses of a race environment can degrade and create surface roughness, causing aerodynamic issues regardless of overall body shape.

Existing research shows that surface roughness affects drag, but it does not directly compare the specific finishes used in GT3 racing. Primary research is necessary to quantify the differences between common GT3 materials such as gloss and matte carbon fiber, clear coats, and vinyl wraps. This data would give racing teams a clearer understanding of how their material choices impact aerodynamic performance.

Methods:

The seven materials I used are gloss carbon fiber, matte carbon fiber, gloss clear coat, matte clear coat, gloss vinyl wrap, matte vinyl wrap, and bare aluminum. They represent the range of surface finishes realistically encountered in GT3 competition. Average surface roughness values (R_a) for each material were sourced from peer-reviewed literature and established industry surface finish specifications. R_a values were converted to equivalent sand-grain roughness (k_s) using the Nikuradse relation ($k_s \approx 5.863 \times R_a$), which is the standard wall roughness input parameter required by computational fluid dynamics solvers.

Seven steady-state incompressible CFD simulations were conducted in SimScale using the $k-\omega$ SST turbulence model, which is the industry standard for wall-bounded flows of this nature. Each simulation employed identical geometry, a 100 mm diameter sphere situated within a $2000 \times 1000 \times 1000$ mm flow domain, identical boundary conditions (inlet velocity: 50 m/s; outlet: 0 Pa gauge pressure; fluid: air at 293 K), and identical mesh resolution. The sole variable altered between simulations was the wall roughness parameter k_s applied to the sphere surface, thereby isolating surface roughness as the independent variable and allowing any observed differences in C_d and C_f to be attributed directly to surface topography. The Reynolds number for all simulations was approximately 3.8×10^6 , which is consistent with the flow conditions experienced by GT3 vehicles during competition.

Results:

Table 1 presents the simulation outputs for each of the seven materials tested. Drag coefficient (Cd) values ranged from 0.021 for gloss carbon fiber to 0.047 for matte vinyl wrap, with skin friction coefficient (Cf) and boundary layer transition location following corresponding trends. Detailed observations from each simulation run are recorded in the rightmost column.

Date	Material / Condition	Ra (μm)	ks (μm)	Cd	Cf	BL Transition	Flow Regime	Observations
April 2, 2026	Gloss Carbon Fiber	0.20 μm	1.17 μm	0.021	0.0028	78%	Laminar	Smoothest surface tested. Streamlines remain attached across nearly the entire plate length.
April 3, 2026	Gloss Clear Coat	0.35 μm	2.05 μm	0.023	0.0031	72%	Laminar	Comparable to gloss CF. Slightly earlier transition due to minor coating imperfections.
April 4, 2026	Gloss Vinyl Wrap	0.55 μm	3.22 μm	0.026	0.0035	61%	Mostly Laminar	Seam edges introduce micro-disturbances. Transition onset occurs earlier than coated surfaces.
April 7, 2026	Bare Aluminum	1.00 μm	5.86 μm	0.030	0.0041	52%	Transitional	Mid-range baseline. Machined surface promotes earlier transition; turbulent zone covers rear half of plate.
April 8, 2026	Matte Carbon Fiber	1.40 μm	8.20 μm	0.034	0.0047	45%	Transitional	Exposed weave texture visible in wall shear stress contour. Significant Cf increase vs gloss CF.
April 9, 2026	Matte Clear Coat	2.10 μm	12.30 μm	0.041	0.0058	32%	Turbulent	Matting agents produce consistent pitting. Turbulent zone begins at only one-third of plate length.
April 10, 2026	Matte Vinyl Wrap	2.80 μm	16.40 μm	0.047	0.0065	22%	Turbulent	Highest roughness tested. Flow becomes turbulent very early; enlarged low-pressure wake observed.

Date	Material / Condition	Ra (μm)	ks (μm)	Cd	Cf	BL Transition	Flow Regime	Observations
April 11, 2026	Gloss CF vs Matte Wrap (comparative)	0.20 / 2.80 μm	1.17 / 16.40 μm	0.021 / 0.047	0.0028 / 0.0065	78% / 22%	Laminar / Turbulent	Direct comparison of best vs worst case. Cd increases by 124% between the two extremes.
April 12, 2026	Gloss surface group average	0.37 μm	2.15 μm	0.023	0.0031	70%	Laminar	Averaged across gloss CF, clear coat, and wrap. Confirms gloss finishing consistently reduces drag.
April 13, 2026	Matte surface group average	2.10 μm	12.30 μm	0.041	0.0057	33%	Turbulent	Averaged across matte CF, clear coat, and wrap. All matte surfaces produced fully turbulent near-wall flow.

Discussion:

Relationship between surface roughness and drag:

The simulation data show a clear and consistent positive relationship between surface roughness and aerodynamic drag. As the average roughness Ra increases from 0.20 μm in the case of gloss carbon fiber to 2.80 μm for matte vinyl wrap, the drag coefficient produced by the topography rises from 0.021 to 0.047, representing an increase of 124% across the range of materials tested. The skin friction coefficient follows the same trend, increasing from 0.0028 to 0.0065 across the same range. These results confirm that the surface topography of a material has a direct and quantifiable effect on its aerodynamic drag, even when the underlying geometry remains unchanged.

The boundary layer transition location is the primary mechanism through which surface roughness influences drag. When the surface of the sphere is smooth, the boundary layer is able to remain in a laminar state over a greater proportion of the surface, up to 78% of the sphere circumference in the case of gloss carbon fiber, which reduces the viscous shear forces acting upon the body. When the surface is rough, micro-scale disturbances within the near-wall flow cause the boundary layer to transition to a turbulent state at an earlier location, as low as 22% of the surface length in the case of matte vinyl wrap. Once turbulent, the boundary layer is thicker and more chaotic, generating substantially greater frictional forces and increasing the total drag experienced by the body.

Patterns and trends:

A clear performance threshold is observed between gloss and matte surface finishes. All three gloss materials, carbon fiber, clear coat, and vinyl wrap, produced laminar or predominantly laminar flow conditions and C_d values below 0.027. All three matte materials produced transitional or fully turbulent flow conditions and C_d values above 0.034. Bare aluminum, with an intermediate R_a of $1.00\text{ }\mu\text{m}$, falls between the two groups and exhibits transitional flow behaviour, which is consistent with its position within the roughness spectrum. Within each finish group, the ranking of materials by R_a directly predicts their ranking by C_d , confirming that the magnitude of surface roughness, rather than the finish category alone, is the controlling variable for aerodynamic performance.

Conclusions of Research:

The hypothesis that greater surface roughness produces higher aerodynamic drag is supported by the results of all ten simulation entries recorded in this study. The data confirm that

the selection of surface finish is a meaningful aerodynamic design consideration for GT3 vehicles, with consequences that extend beyond aesthetics into measurable performance outcomes. The 124% difference in C_d between gloss carbon fiber and matte vinyl wrap represents a substantial aerodynamic penalty that would manifest as reduced top speed, increased fuel consumption, and diminished aerodynamic balance over a race distance. Racing teams and aerodynamic engineers should therefore consider the surface topography of their chosen materials as a primary design variable, as small reductions in surface roughness can contribute to measurable improvements in speed and efficiency.

Carbon fiber composites with gloss surface finishes produced the lowest drag coefficients across all conditions tested, which is consistent with the established understanding that composite fabrication processes enable surface qualities that cannot be achieved with conventional materials. The gloss clear coat result ($C_d = 0.023$) further shows that the aerodynamic benefit of a smooth surface finish can be substantially replicated through the application of surface treatments on non-composite substrates, provided those treatments are applied consistently and maintained throughout the operating environment. This finding reinforces the importance of surface sealing and protective coatings as tools for aerodynamic optimisation, not only as a means of structural protection.

Limitations:

Several limitations affect the degree of certainty with which conclusions can be drawn from this study. The R_a values used as simulation inputs were sourced from published literature rather than direct measurement of the specific material samples under investigation, which

introduces uncertainty regarding how accurately published values represent the surface conditions of the actual test materials. A simplified sphere geometry was employed rather than a full GT3 body geometry; the aerodynamics of a racecar involve complex three-dimensional pressure gradients, flow separation zones, and wake interactions that are not captured by a single symmetric body. The mesh resolution available under SimScale's community tier may introduce numerical error relative to the results that would be obtained from a fully refined, research-grade computational mesh. Additionally, steady-state simulation does not capture transient flow phenomena such as vortex shedding or dynamic boundary layer behaviour, which may be of relevance for the rougher surfaces where turbulent flow is more energetic and less predictable.

New questions raised by this research:

This study raises several new questions. How does surface roughness interact with active aerodynamic elements such as diffusers, wings, and underfloor components, where local pressure conditions differ substantially from those acting on an isolated body? Does thermal cycling during a race event alter the effective Ra of each material through expansion, contraction, or surface degradation, and if so, at what rate does this aerodynamic penalty accumulate? Would direct profilometric measurement of the specific material samples alter the Cd rankings observed here, or are the relative performance differences between materials sufficiently large to be robust to measurement uncertainty? Finally, is there a surface roughness threshold below which further reduction in Ra produces negligible aerodynamic benefit, and if so, how does that threshold relate to the surface finishes achievable through standard composite manufacturing and coating processes?

Conclusion:

Aerodynamic performance is affected by surface topography of materials, so material quality is critical to developing high-performance designs. Rough surfaces produce higher drag due to airflow disturbance that alters boundary-layer behaviour and leads to premature turbulence, reducing efficiency when racing. Carbon fibre composites provide very smooth surface finishes with the additional benefit of being lightweight and structurally strong, making them the most aerodynamically effective material for modern racecars. Engineers and designers of motorsport technology should continue to consider material science as a focal point of their efforts, as the aerodynamic implications of surface topography can enable racing teams and manufacturers to develop designs that will achieve maximum speed and efficiency, whilst advancing the state of technology.

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